

Kimberly Wang

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EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S. Electrical Engineering, GPA: 3.71

Aug 2024 - Expected Dec 2027

Relevant Coursework: Electric Circuits, Computer Components and Operations, Embedded Control, Intro to Electronics (Spring 2026), Intro to Engineering Design (Spring 2026), Signals and Systems (Spring 2026)

TECHNICAL SKILLS

Hardware: Arduino, MSP432, ESP32, Basys 3, Oscilloscope, Waveform Generator, Spectrum Analyzer, Multimeter

Software: LTspice, Cadence, Altium Designer, Xilinx Vivado, SolidWorks, LabVIEW, MATLAB, Simulink, Office Suite

Languages: Python, MicroPython, C/C++, SQL, VHDL, MATLAB

EXTRACURRICULARS

Undergraduate Researcher @ RPI Mercer XLab

Jan 2025 - Present

Rensselaer Polytechnic Institute

Troy, NY

Software Configured Breadboard (SCB) Supervised by ADI Fellow Douglas Mercer and Prof. Shayla Sawyer

- Used Python and ALICE software to computerize the wiring of breadboards to reduce debugging and wiring issues
- Executed board-level validation and root-cause analysis utilizing automated Python scripts and a multimeter
- Contributed to hardware design reviews, providing feedback on the PCB layout and component choices
- Authored technical documentation and user guides to streamline the onboarding process for new users
- Collaborating with RPI's AI club to utilize Cursor AI to automate circuit design, building, and testing

Undergraduate Student Assistant for "Intro to ECSE"

Aug 2025 - Dec 2025

Rensselaer Polytechnic Institute

Troy, NY

- Help students in debugging and implementing C programming on Arduino and MATLAB
- Advise on analog and digital circuit building, testing, and measurement
- Clarify questions about theoretical concepts within contexts of labs

PROJECTS

Desktop Buddy | *Altium Designer, Benchtop Equipment, ESP32, C/C++, Autodesk Fusion*

Dec 2025 - Present

- Designed custom hardware and software of a desktop robot based on an ESP32 Microcontroller
- Currently in process of implementation

CO Sensor System | *LTspice, Benchtop Equipment, Analog Circuit*

Sep 2025 - Dec 2025

- Created a multi-stage analog system for maintaining safety from CO in buildings
- Designed and tested complete analog circuit for indication, ventilating, and alarming systems
- Performed failure analysis and problem solving to ensure functionality of physical circuit

Compass Controlled TI RSLK Car | *MSP432, C, I2C*

Apr 2025

- Designed a closed-loop navigation system using a magnetometer via I2C protocols for real time directions
- Implemented PID controls in C to minimize errors in motor movements
- Configured GPIO interrupts to handle data acquisition and maintain stable motor control

Implementation of ALU to Demonstrate Boolean Functions | *Vivado, VHDL, FPGA (Basys 3)*

Feb 2025

- Authored VHDL code to write logic for an ALU with 4 bits for an adder, subtractor, bit parity, and comparator
- Used Vivado to define constraints and generate bitstream to be sent to Basys 3 FPGA board
- Tested functions with on-board switches as inputs and LEDs for outputs

Puzzle Solving Autonomous Raspberry Pi-Based Cart | *Python, Raspberry Pi*

Dec 2024

- Designed a strategy to determine a path based on input on adjacent grids to reach an unknown target grid coordinate and travel back to the starting position
- Program the cart in Python to travel within a physical 4x4 grid with sensor inputs and motor control

PI Controller Design | *LTspice, Oscilloscope*

Nov 2024

- Designed and simulated an analog PI controller in LTspice to achieve a steady state voltage in 4s
- Verified the design by building and testing the circuit to get results with < 5% error from simulation